

Diffusion-Based Saliency Map Generation from RGB Images

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Abstract

We propose SalFusion, a diffusion-based model for generating dense visual saliency maps from RGB images. The model combines a U-Net backbone with spatial self-attention and cross-attention, trained with a denoising diffusion objective and auxiliary reconstruction losses (L1, MSE, SSIM). Classifier-free guidance enables a controllable sharpness–structure trade-off during sampling, and a faster deterministic sampler is also supported. On SALICON, our results are competitive on standard saliency metrics, with particularly strong gains in AUC-Judd and NSS, while the lower CC highlights a trade-off between localization accuracy and global density matching.

Keywords: Visual saliency, diffusion models, DDPM, DDIM, classifier-free guidance

1 Introduction

Visual saliency prediction aims to estimate where humans are likely to fixate when viewing a scene. Early models relied on hand-crafted centre–surround features [1], while modern approaches use deep neural networks to learn hierarchical representations directly from fixation data, significantly improving performance on benchmark datasets. Diffusion models, particularly Denoising Diffusion Probabilistic Models (DDPMs) [2], have recently emerged as stable and expressive generative models. Trained via a simple noise-prediction objective rather than adversarial learning, diffusion models iteratively refine noisy samples into structured outputs, making them well-suited for dense prediction tasks that require both global coherence and local detail. Their success in other structured vision problems motivates their use for saliency estimation.

In this work, we formulate visual saliency prediction as a conditional diffusion process that generates a normalized single-channel saliency map from an RGB image. Our model learns the reverse denoising process of a DDPM conditioned on image features through both early fusion and cross-attention mechanisms. This iterative generative formulation enables the model to capture global scene context and produce structurally coherent saliency maps. We evaluate the approach on the SALICON and MIT1003 datasets and analyze the impact of attention mechanisms, classifier-free guidance, and sampling strategies on saliency quality.

1.1 Main Contributions

Our contributions are threefold. First, we introduce a conditional denoising diffusion framework for visual saliency estimation, modelling saliency maps as samples from a learned generative process rather than deterministic regression outputs. Second, we design a U-Net–based DDPM augmented with spatial self-attention and cross-attention, and apply classifier-free guidance during inference to produce sharper and more localized saliency predictions. Third, we benchmark the proposed model on SALICON and MIT1003 using standard saliency metrics, and we conduct ablation studies on attention modules, guidance, and sampling strategies to highlight the main performance trade-offs.

Diffusion models offer an iterative denoising formulation that aggregates multi-scale context while preserving fine detail. When conditioned on image features and combined with classifier-free guidance, they provide

an explicit mechanism to trade off sharpness and structural fidelity, which is particularly useful for saliency where precise localization of fixation peaks matters. This is a practical advantage over purely feed-forward predictors, because the sampler can be tuned to favor either sharper localization or smoother density structure. This iterative, generative viewpoint also enables sampling-based uncertainty estimates and multiple plausible saliency predictions for ambiguous scenes.

2 Related Work

2.1 CNN-Based Saliency Prediction

Early deep learning approaches to visual saliency were dominated by convolutional neural networks (CNNs). These models used hierarchical feature representation to approximate human fixation. Classical architectures such as MLNet and SalNet proved that CNN-based saliency prediction was viable. Later models improved upon these results by introducing multi-scale feature aggregation and improving spatial reasoning. Models such as SimpleNet [3], MDNSal [3], and MSI-Net [4] made further improvements using dilated convolutions, multi-branch encoders, and probabilistic modeling. Despite strong performance, CNN-based models are limited by their inherently local receptive fields, restricting their ability to model long-range dependencies.

2.2 GAN-Based Models

Following the success of generative adversarial networks (GANs) [5] in learning data distributions through adversarial training, saliency prediction has also benefited from applying this technique to saliency map prediction. SalGAN [6] uses this framework where a generator produces saliency given an image and a discriminator distinguishes generated vs GT saliency maps. The architecture consists of a VGG-16-based encoder-decoder generator paired with a discriminator trained to distinguish ground-truth saliency maps from predicted ones. The generator is optimized with a combination of a binary cross-entropy (BCE) content loss and an adversarial loss imposed by the discriminator. SalGAN was evaluated with saliency-specific metrics (CC, NSS, AUC-Judd) on SALICON and MIT benchmarks. SalGAN achieved competitive scores compared to state-of-the-art CNN-based saliency models at the time, outperforming traditional regression networks trained solely with pixel-wise losses.

2.3 Recent Work

While CNN and GAN-based approaches generate saliency maps in a single forward pass, the use of attention mechanisms enables other models to iteratively refine saliency maps over multiple timesteps. This allows the model to accumulate spatial context, correct earlier predictions and gradually refine their focus to the most important parts of the image. The Saliency Attentive Model (SAM) [7] is a family of models that adopted this technique. There are two variations of this model, SAM-VGG and SAM-ResNet, which use VGG-16 and ResNet-50 backbones respectively. SAM starts with a learned understanding of where humans usually look in images, and then uses an attention-based recurrent module to repeatedly focus on important regions, refining the saliency map step by step., leading to substantial gains over earlier CNN and GAN-based approaches. On the SALICON validation set, SAM achieves strong performance, with SAM-ResNet in particular delivering high NSS and CC scores due to its deeper feature representation and recurrent refinement process. Vision-language based approaches to saliency prediction have gained popularity, leveraging recent advances in integrating large language and vision models. Recent multimodal and transformer-based methods such as ALBEF [8], BLIP [9], ImageBind [10], and VLSal [11] show that semantic and cross-modal representations can improve saliency prediction, especially for complex real-world scenes. Our work is complementary to this line of research: we study whether an iterative generative formulation can provide competitive localization performance without relying on large pretrained multimodal backbones.

3 Method

3.1 Diffusion Models

DDPMs [2] learn to reverse a forward Gaussian noising process via noise prediction. DDIM [12] enables deterministic or fast sampling, while classifier-free guidance [13] improves conditional fidelity. Diffusion has shown strong performance in dense prediction tasks such as segmentation and depth estimation.

3.2 Saliency Prediction

Earlier saliency modelling methods relied on handcrafted features [1]. With deep learning, CNN-based regressors dominated until adversarial methods such as SalGAN [6] introduced GAN losses to encourage realistic saliency distributions.

3.3 U-Net Architectures

U-Net [14] architectures remain foundational for dense prediction. Residual connections [15], GroupNorm [16], and spatial attention modules further enhance representational power. We employ a U-Net with cross-attention to integrate image context.

3.4 Model Architecture

We adopt a conditional U-Net backbone tailored for diffusion-based dense prediction. The network takes as input a 4-channel tensor formed by concatenating the RGB image with the noisy saliency map at diffusion timestep t . A sinusoidal timestep embedding is computed and projected through a two-layer MLP to a 128-dimensional vector, which is injected into every residual block via additive conditioning. This allows the network to adapt its denoising behaviour across timesteps rather than learning a single, timestep-agnostic mapping.

The U-Net follows a symmetric encoder-decoder structure with three downsampling stages and corresponding upsampling stages. The encoder progressively increases channel capacity from 64 to 128 and 256 channels using residual blocks, followed by a bottleneck operating at 512 channels. Each stage consists of a residual convolutional block with Group Normalization and SiLU activation, followed by spatial downsampling. The decoder mirrors this structure using transposed convolutions and skip connections to fuse encoder features, enabling recovery of fine spatial detail while preserving high-level semantics. The full backbone contains approximately 11.8M trainable parameters.

To enhance global reasoning, we incorporate spatial self-attention at the bottleneck, where the feature maps have the lowest spatial resolution and highest semantic abstraction. This module uses multi-head attention to model long-range dependencies between distant regions, which is important for saliency prediction in cluttered scenes with multiple competing objects. In parallel, a lightweight RGB context encoder processes the clean image through three convolutional layers (128, 256, and 512 channels) to produce features aligned with the bottleneck resolution. These features are injected via cross-attention, enabling the denoising pathway to align saliency generation with high-level semantic cues rather than relying solely on early-layer concatenation.

The decoder fuses the attention-enhanced bottleneck representation with encoder skip features to reconstruct a full-resolution output. A final 1×1 convolution projects the last 64-channel feature map to a single channel corresponding to the predicted noise component. This prediction is used by the diffusion sampler to iteratively recover the clean saliency map. Overall, the architecture combines multi-scale feature fusion, timestep-aware conditioning, and attention-based image guidance to balance spatial precision with global contextual understanding during the reverse diffusion process. A structured summary of the network configuration is provided in Table 1.

Table 1: Summary of the conditional U-Net architecture used in the diffusion model. Resolutions are given for input size 256×192 .

Stage	Resolution	Channels	Block Type
Input	256×192	4	Residual Block
Encoder 1	128×96	64	Residual Block + MaxPool
Encoder 2	64×48	128	Residual Block + MaxPool
Encoder 3	32×24	256	Residual Block + MaxPool
Bottleneck	16×12	512	Residual Block + Self-Attn + Cross-Attn
Decoder 3	32×24	256	Transposed Conv + Residual Block (skip)
Decoder 2	64×48	128	Transposed Conv + Residual Block (skip)
Decoder 1	128×96	64	Transposed Conv + Residual Block (skip)
Output Head	256×192	1	1×1 Conv (Noise Prediction)

3.5 Loss Function

Training uses a multi-component objective that combines the standard DDPM noise-prediction loss with auxiliary reconstruction and perceptual structure terms. The auxiliary terms provide a more direct supervised signal on the predicted clean saliency map and help reduce over-smoothing, which can otherwise blur fixation peaks. For clarity, all losses are computed on saliency maps normalized to $[0, 1]$ and weighted to operate on comparable scales. The overall objective is:

$$\mathcal{L}_{\text{total}} = \lambda_{\varepsilon} \mathcal{L}_{\varepsilon} + \lambda_1 \mathcal{L}_1 + \lambda_2 \mathcal{L}_2 + \lambda_s \mathcal{L}_{\text{SSIM}}, \quad (1)$$

where we set $\lambda_{\varepsilon} = 1.0$, $\lambda_1 = 0.5$, $\lambda_2 = 0.5$, and $\lambda_s = 0.5$.

The noise-prediction term follows the DDPM objective. We sample a timestep $t \sim \mathcal{U}\{1, \dots, T\}$ and construct a noisy target saliency map $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\varepsilon}$, where $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and $\bar{\alpha}_t$ is the cumulative product of the noise schedule. The model predicts the injected noise conditioned on the RGB input $\mathbf{c}$, and we minimize:

$$\mathcal{L}_{\varepsilon} = \mathbb{E}_{\mathbf{x}_0, t, \boldsymbol{\varepsilon}} \|\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}_{\theta}(\mathbf{x}_t, t, \mathbf{c})\|_2^2. \quad (2)$$

This term is the core objective enabling accurate reverse diffusion.

To encourage spatial fidelity and well-calibrated intensities, we additionally supervise the estimated clean saliency map $\hat{\mathbf{x}}_0$ with pixel-wise losses:

$$\mathcal{L}_1 = \|\hat{\mathbf{x}}_0 - \mathbf{x}_0\|_1, \quad \mathcal{L}_2 = \|\hat{\mathbf{x}}_0 - \mathbf{x}_0\|_2^2. \quad (3)$$

The L1 term promotes sharper peaks and reduces blur, while the L2 term provides smooth gradients and stabilizes optimization by penalizing large deviations more strongly.

Because saliency maps are smooth density fields where perceptual structure matters, we also incorporate a lightweight SSIM-based loss:

$$\mathcal{L}_{\text{SSIM}} = 1 - \text{SSIM}(\hat{\mathbf{x}}_0, \mathbf{x}_0), \quad (4)$$

which encourages agreement in local luminance, contrast, and spatial structure beyond pixel-wise matching. In combination, these terms balance accurate denoising with structural fidelity, improving both localization of fixation peaks and global map coherence.

3.6 Training and Inference

Training uses AdamW optimizer with learning rate $1e-4$, batch size 8, timesteps 500, and cosine annealing noise scheduling over 60 epochs. Training took ~ 3 hours on a NVIDIA RTX A4500 GPU. Mixed precision (AMP) accelerates training.

Compute: Training on SALICON with batch size 8 required approximately 3 hours on a single NVIDIA RTX A4500. For inference we report timings for both full DDPM sampling and a faster DDIM sampler (50

steps) as a practical trade-off: DDIM reduces sampling cost substantially while retaining most accuracy. Practical deployment options to reduce cost include timestep reduction, denoising distillation, or a lightweight deterministic refinement head.

At inference, we use DDPM sampling for best quality, as the full stochastic reverse process better matches the training objective and yields more refined spatial structure. We optionally employ DDIM for $\sim 10\times$ faster sampling due to the reduced number of denoising steps, at the cost of some loss in fine detail and localization accuracy. Classifier-free guidance with scale $s = 1.5$ is applied to strengthen conditioning on the RGB image. By amplifying the difference between conditional and unconditional predictions, guidance encourages sharper activations around likely fixation regions and improves fixation localization, though overly large guidance scales were observed to produce overconfident or overly peaked saliency maps.

4 Results

4.1 Dataset and Preprocessing

We train on the SALICON dataset [17], which provides MS-COCO images with mouse-tracking fixation data. The dataset contains 10k training images, 5k validation images, and 5k test images, with no fixation annotations available for the test split.

Evaluation protocol: our outputs are resized to 256×192 and we apply Gaussian smoothing ($\sigma = 8$) prior to metric computation. For the comparison table, prior-method values are reported from their published results, while SalFusion entries are our own results computed using this evaluation protocol. We report single-run results in this submission, so no confidence intervals are included.

Fixation JSONs are converted into continuous saliency maps using Gaussian smoothing ($\sigma = 8$), normalized into $[0, 1]$, and resized to 256×192 .

4.2 Evaluation Metrics

We evaluate model performance using three standard saliency prediction metrics computed on the validation set. CC measures the linear correlation between the predicted saliency map and the ground-truth human saliency map, NSS evaluates the average normalized saliency value at fixation points, and AUC-Judd measures how well the model ranks human fixations above non-fixated pixels across the image.

4.3 Quantitative Results

Table 2 presents the quantitative comparison between our model, SalFusion, and other models on the SALICON (validation) and MIT1003 datasets. The values for prior methods are reported in [11], while the SalFusion entries are our own results under the evaluation protocol described above.

Based on these results, our diffusion model is stronger on the AUC and NSS metrics, but weaker on CC. This reflects a trade-off between discriminative ranking performance and pixelwise structural fidelity. The large gains in AUC-Judd and NSS indicate that the model is more effective at ranking salient versus non-salient regions and at placing high values near fixation locations, while the lower CC suggests that the predicted maps are less linearly aligned with the full smoothed density structure of the ground truth. Overall, our diffusion approach produces saliency maps that are more effective for localization and detection-style evaluation but less accurate in reproducing the full spatial density structure captured by CC.

Limitations and calibration: The lower CC scores reflect reduced per-pixel linear agreement with smoothed ground-truth densities, caused by the model’s tendency to produce sharper, peakier maps. This is a practical limitation of the current formulation, particularly for applications that require faithful global density matching rather than strong fixation localization. We find that light post-hoc Gaussian smoothing or a small learned intensity-calibration stage can substantially increase CC with only minor reductions in NSS/AUC. Diffusion models are also more computationally expensive at inference than single-pass feed-forward baselines, so the

Table 2: Quantitative comparison of models trained on SALICON train and evaluated on SALICON validation and MIT1003. Prior-method values are reported in [11]; SalFusion entries are our own results.

SALICON (Validation)				MIT1003			
Model	NSS $\uparrow$	CC $\uparrow$	AUC-J $\uparrow$	Model	NSS $\uparrow$	CC $\uparrow$	AUC-J $\uparrow$
SALICON	1.5003	0.7486	0.8304	SALICON	1.6648	0.5012	0.8380
MLNet	1.5021	0.6746	0.8094	MLNet	1.7424	0.4836	0.8241
SalGAN	1.7989	0.8601	0.8601	SalGAN	2.1328	0.6245	0.8779
SAM-VGG	1.7637	0.8247	0.8524	SAM-VGG	2.0078	0.5788	0.8539
SAM-ResNet	1.8121	0.8398	0.8543	SAM-ResNet	2.1320	0.6153	0.8694
TranSalNet	1.7996	0.8325	0.8496	TranSalNet	1.7829	0.5520	0.8693
TempSAL	1.7656	0.8245	0.8395	TempSAL	1.7650	0.5203	0.8497
GSGNet	1.8339	0.8466	0.8548	GSGNet	1.9846	0.5795	0.8656
TGSal	1.8606	0.8816	0.8658	TGSal	2.2856	0.6632	0.8873
ALBEF	1.7368	0.8469	0.8564	ALBEF	1.9971	0.5958	0.8743
BLIP	1.8070	0.8649	0.8603	BLIP	2.0964	0.6172	0.8768
ImageBind	1.7248	0.8423	0.8558	ImageBind	1.9851	0.5925	0.8728
VLSal	1.9492	0.9085	0.8718	VLSal	2.4658	0.7088	0.8986
SalFusion	2.6650	0.7245	0.9291	SalFusion	2.5327	0.4855	0.8925

current work prioritizes quality and controllability over sampling efficiency. We report these trade-offs explicitly and discuss them further in the supplementary material.

4.4 Qualitative Results

Fig. 1 shows that model captures main fixation regions, but predicts sharper peaks and occasionally misses small secondary regions. It appears that our model favours more confident predictions over matching the per-pixel intensity of the ground-truth saliency maps.



Figure 1: Representative qualitative examples. Each row shows an RGB image, the ground-truth saliency map, and the predicted saliency map from our model.

4.5 Ablation Study

We conduct an ablation study to assess the impact of toggling features on / off on model performance. The results of our study captured in Table 3 suggest that guidance is the most influential factor, while DDIM harms accuracy in our setup.

Table 3: Ablation Results (CFG = classifier-free guidance).

CFG	DDIM	SA	CA	AUC-J	NSS	CC
1.5	Off	T	T	0.93	2.62	0.7185
0.0	Off	T	T	0.91	2.257	0.5895
1.5	On	T	T	0.86	1.58	0.42
1.5	Off	F	F	0.89	2.42	0.639

5 Conclusion

Our research and experiments show that diffusion models show competitive performance relative to GANs for saliency prediction. Despite a slight underperformance in CC, our diffusion model outperformed other models in other key areas. The stronger CC score of GAN-based approaches is due to its adversarial training, which encourages generators to match the global distribution of saliency maps. Diffusion models prioritise producing high-confidence predictions at likely fixation points, which boosts metrics such as AUC and NSS. This behaviour is consistent with our qualitative analysis, where saliency concentrated sharply around high-confidence regions.

Beyond GANs, our results also demonstrate that diffusion models compare favourably to traditional CNN-based saliency networks such as SimpleNet and MDNSal, achieving higher AUC and NSS while offering more stable training and better spatial coherence. Compared to recurrent attention models like SAM-VGG and SAM-ResNet, diffusion models avoid the complexity and computational overhead of multi-step refinement while still delivering competitive localisation accuracy. SAM achieves strong NSS through iterative ConvLSTM refinement, but diffusion models provide a simpler, more general framework that excels at precise fixation localisation.

Overall, whether a diffusion model or a GAN is more suitable depends on the application. Diffusion models may be more reliable in scenarios where accurate, sharply localised fixation prediction is essential, whereas GAN-based approaches may be preferable when matching global saliency statistics is the primary objective. For saliency prediction, the main advantage of diffusion is the ability to model structured spatial outputs while providing an explicit control over sharpness and structural fidelity through guidance.

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